

# Integration Review

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## 1 Problem 1

### 1.1 Problem

$$\int x \ln(x) dx$$

### 1.2 Solution

$$\int x \ln(x) dx = \frac{x^2 \ln(x)}{2} - \frac{1}{2} \int \frac{x^2}{x} dx$$

$$\int x \ln(x) dx = \frac{x^2 \ln(x)}{2} - \frac{x^2}{4} + C$$

## 2 Problem 2

$$\int x^2 e^x dx$$

### 2.1 Problem

$$\int x^2 e^x dx$$

### 2.2 Solution

$$\int x^2 e^x dx = e^x x^2 - \int 2x e^x dx$$

$$\int x^2 e^x dx = e^x x^2 - (2x e^x - \int 2e^x dx)$$

$$\int x^2 e^x dx = e^x x^2 - 2x e^x + 2e^x + C$$

### 3 Problem 3

#### 3.1 Problem

$$\int x^2 \sin(x) dx$$

#### 3.2 Solution

$$\begin{aligned}\int x^2 \sin(x) dx &= -x^2 \cos(x) + \int 2x \cos(x) dx \\ \int x^2 \sin(x) dx &= -x^2 \cos(x) + (2x \sin(x) - \int 2 \sin(x) dx) \\ \int x^2 \sin(x) dx &= -x^2 \cos(x) + 2x \sin(x) + 2 \cos(x) + C\end{aligned}$$

### 4 Problem 4

#### 4.1 Problem

$$\int e^x \sin(x) dx$$

#### 4.2 Solution

$$\begin{aligned}\int e^x \sin(x) dx &= e^x \sin(x) - \int e^x \cos(x) dx \\ \int e^x \sin(x) dx &= e^x \sin(x) - (e^x \cos(x) + \int e^x \sin(x) dx) \\ 2 \int e^x \sin(x) dx &= e^x \sin(x) - e^x \cos(x) \\ \int e^x \sin(x) dx &= \frac{e^x \sin(x) - e^x \cos(x)}{2} + C\end{aligned}$$

### 5 Problem 5

#### 5.1 Problem

$$\int \cos(\ln(x)) dx$$

#### 5.2 Solution

$$\begin{aligned}\int \cos(\ln(x)) dx &= x \cos(\ln(x)) + \int \frac{x \sin(\ln(x))}{x} dx \\ \int \cos(\ln(x)) dx &= \frac{1}{2} (x \sin(\ln(x)) + x \cos(\ln(x))) + C\end{aligned}$$

## 6 Problem 6

### 6.1 Problem

$$\int \frac{1}{x^2(x^2+1)} dx$$

### 6.2 Solution

$$\begin{aligned} \int \frac{1}{x^2(x^2+1)} dx &= \int \frac{A}{x} + \frac{B}{x^2} + \frac{Cx+D}{x^2+1} dx \\ 1 &= A(x^3+x) + B(x^2+1) + C(x^3) + D(x^2) \end{aligned}$$

$$A = 0, \quad B = 1, \quad C = 0, \quad D = -1$$

$$\begin{aligned} \int \frac{0}{x} + \frac{1}{x^2} + \frac{(0)x + (-1)}{x^2+1} dx \\ \frac{-1}{x} - \tan^{-1}(x) + C \end{aligned}$$

## 7 Problem 7

### 7.1 Problem

$$\int \frac{3x+2}{(x-2)^2(x^2+1)} dx$$

### 7.2 Solution

$$\begin{aligned} \int \frac{3x+2}{(x-2)^2(x^2+1)} dx &= \int \frac{A}{x-2} + \frac{B}{(x-2)^2} + \frac{Cx+D}{x^2+1} dx \\ 3x+2 &= A(x-2)(x^2+1) + B(x^2+1) + (Cx+D)(x-2)^2 \\ 3x+2 &= A(x^3-2x^2+x-2) + B(x^2+1) + C(x^3-4x^2+4x) + D(x^2-4x+4) \\ A &= \frac{-17}{25}, \quad B = \frac{8}{5}, \quad C = \frac{17}{25}, \quad D = \frac{-6}{25} \\ \int \frac{-17/25}{(x-2)} + \frac{8/5}{(x-2)^2} + \frac{(17/25)x + (-6/25)}{(x^2+1)} dx \\ \frac{-17}{25} \ln|x-2| - \frac{8}{5} \frac{1}{x-2} + \frac{17}{50} \ln(x^2+1) - \frac{6}{25} \tan^{-1}(x) + C \end{aligned}$$

## 8 Problem 8

### 8.1 Problem

$$\int \frac{x-3}{x(x^3-27)} dx$$

## 8.2 Solution

$$\begin{aligned}\int \frac{x-3}{x(x^3-27)} dx &= \int \frac{A}{x} + \frac{Bx+C}{x^2+3x+9} dx \\ x-3 &= A(x^2+3x+9) + B(x^2) + C(x) \\ A &= \frac{1}{9}, \quad B = \frac{-1}{9}, \quad C = \frac{-1}{3} \\ \int \frac{1}{9x} + \frac{(-1/9)x + (-1/3)}{x^2+3x+9} dx \\ \frac{1}{9} \ln|x| - \frac{1}{18} \ln(x^2+3x+9) - \frac{1}{9\sqrt{3}} \tan^{-1}\left(\frac{2x+3}{3\sqrt{3}}\right) + C\end{aligned}$$

## 9 Problem 9

### 9.1 Problem

$$\int \frac{4}{x^2+5x-14} dx$$

### 9.2 Solution

$$\begin{aligned}\int \frac{4}{x^2+5x-14} dx &= \int \frac{A}{x+7} + \frac{B}{x-2} dx \\ 4 &= A(x-2) + B(x+7) \\ A &= \frac{-4}{9}, \quad B = \frac{4}{9} \\ \int \frac{-4}{9(x+7)} + \frac{4}{9(x-2)} dx \\ \frac{-4}{9} \ln|x+7| + \frac{4}{9} \ln|x-2| + C\end{aligned}$$

## 10 Problem 10

### 10.1 Problem

$$\int \frac{4x-11}{x^3-9x^2} dx$$

## 10.2 Solution

$$\int \frac{4x - 11}{x^3 - 9x^2} dx = \int \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x - 9} dx$$

$$A = \frac{-25}{81}, \quad B = \frac{11}{9}, \quad C = \frac{25}{81}$$

$$\int \frac{-25/81}{x} + \frac{11/9}{x^2} + \frac{25/81}{x - 9} dx$$
$$\frac{-25}{81} \ln|x| - \frac{11}{9x} + \frac{25}{81} \ln|x - 9| + C$$

## 11 Problem 11

### 11.1 Problem

$$\int 2\sin^2(7x) dx$$

### 11.2 Solution

$$\frac{2}{1} \int \frac{1 - \cos(14x)}{2}$$
$$x - \frac{\sin(14x)}{14} + C$$

## 12 Problem 12

### 12.1 Problem

$$\int \frac{5}{13} \cos^2(12x) dx$$

### 12.2 Solution

$$\frac{5}{13} \int \frac{1 + \cos(24x)}{2}$$
$$\frac{5x}{26} + \frac{5\sin(24x)}{624} + C$$

## 13 Problem 13

### 13.1 Problem

$$\int \cos^2(x) \sin^3(x) dx$$

### 13.2 Solution

$$\begin{aligned} & \int \cos^2(x)(1 - \sin^2(x))\sin(x)dx \\ & \int \cos^2(x)\sin(x)dx - \int \cos^4\sin(x)dx \\ & \frac{-\cos^3(x)}{3} + \frac{\cos^5(x)}{5} + C \end{aligned}$$

## 14 Problem 14

### 14.1 Problem

$$\int \frac{2}{8 + x^3}dx$$

### 14.2 Solution

$$\frac{1}{6}\ln|x+2| - \frac{1}{12}\ln(x^2 - 2x + 4) + \frac{1}{2\sqrt{3}}\tan^{-1}\left(\frac{x-1}{\sqrt{3}}\right) + C$$

## 15 Problem 15

### 15.1 Problem

$$\int \frac{14}{\sqrt{7-x}}dx$$

### 15.2 Solution

$$-28\sqrt{7-x} + C$$

## 16 Problem 16

### 16.1 Problem

$$\int \tan(x)dx$$

### 16.2 Solution

$$-\ln|\cos(x)| + C$$

## 17 Problem 17

### 17.1 Problem

$$\int \sin^3(x) dx$$

### 17.2 Solution

$$\begin{aligned} \int (1 - \cos^2(x)) \sin(x) dx \\ -\cos(x) + \frac{\cos^3(x)}{3} + C \end{aligned}$$

## 18 Problem 18

### 18.1 Problem

$$\int e^{7+\cos(x)} \sin(x) dx$$

### 18.2 Solution

$$-e^{7+\cos(x)} + C$$

## 19 Problem 19

### 19.1 Problem

$$\int \frac{e^{1/x}}{x^2} dx$$

### 19.2 Solution

$$-e^{\frac{1}{x}} + C$$

## 20 Problem 20

### 20.1 Problem

$$\int \frac{e^{\ln(2x)}}{3x}$$

### 20.2 Solution

$$\frac{e^{\ln(2x)}}{3} + C$$

## 21 Problem 21

### 21.1 Problem

$$\int \ln(\cos(x))\tan(x)dx$$

### 21.2 Solution

$$\frac{-\ln^2|\cos(x)|}{2} + C$$

## 22 Problem 22

### 22.1 Problem

$$\int \frac{\ln(x)(3 + \ln^2(x))^{1/2}}{x} dx$$

### 22.2 Solution

$$\frac{(3 + \ln^2(x))^{3/2}}{3} + C$$

## 23 Problem 23

### 23.1 Problem

$$\int \sec^2(x)\cot(x)dx$$

### 23.2 Solution

$$\ln|\tan(x)| + C$$

## 24 Problem 24

### 24.1 Problem

$$\int \frac{1}{x^2 + 3x + 3} dx$$

### 24.2 Solution

$$\int \frac{1}{(x + \frac{3}{2})^2 + \frac{3}{4}} dx$$
$$\frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{2x + 3}{\sqrt{3}}\right) + C$$



## 25 Problem 25

### 25.1 Problem

$$\int \frac{1}{x^2 - 2x + 23} dx$$

### 25.2 Solution

$$\int \frac{1}{(x-1)^2 + 22} dx$$
$$\frac{1}{\sqrt{22}} \tan^{-1}\left(\frac{x-1}{\sqrt{22}}\right) + C$$

## 26 Problem 26

### 26.1 Problem

$$\int \frac{2x^3 + 3x^2 - 4x + 8}{(x+3)} dx$$

### 26.2 Solution

After synthetic division...

$$\int 2x^2 - 3x + 5 - \frac{7}{x+3} dx$$
$$\frac{2}{3}x^3 - \frac{3}{2}x^2 + 5x - \ln((x+3)^7) + C$$

## 27 Problem 27

### 27.1 Problem

$$\int \frac{x^4 + x^2}{x-2} dx$$

### 27.2 Solution

After synthetic division...

$$\int x^3 + 2x^2 + 5x + 10 + \frac{20}{x-2} dx$$
$$\frac{x^4}{4} + \frac{2x^3}{3} + \frac{5x^2}{2} + 10x + \ln((x-2)^{20}) + C$$

## 28 Problem 28

### 28.1 Problem

$$\int \frac{x^3 - 1}{x - 1}$$

### 28.2 Solution

After synthetic division...

$$\int (x^2 + x + 1)dx$$

$$\frac{x^3}{3} + \frac{x^2}{2} + x + C$$

## 29 Problem 30

### 29.1 Problem

$$\int e^{3x} \cos\left(\frac{x}{7}\right) dx$$

### 29.2 Solution

$$\frac{147e^{3x} \cos\left(\frac{x}{7}\right)}{442} + \frac{7e^{3x} \sin\left(\frac{x}{7}\right)}{442} + C$$

## 30 Problem 31

### 30.1 Problem

$$\int x^3 \cos(2x^4) dx$$

### 30.2 Solution

$$\frac{\sin(2x^4)}{8} + C$$

## 31 Problem 32

### 31.1 Problem

$$\int (3x^3 + 2x^2 - 7x)(e^x) dx$$

### 31.2 Solution

$$(3x^3 - 7x^2 + 7x - 7)e^x + C$$

## 32 Problem 33

### 32.1 Problem

$$\int \sec^4(2x)\tan^2(2x)dx$$

### 32.2 Solution

$$\frac{\tan^3(2x)}{6} + \frac{\tan^5(2x)}{10} + C$$

## 33 Problem 34

### 33.1 Problem

$$\int \frac{1 + 7\sin^3(x)}{\cos^2(x)}dx$$

### 33.2 Solution

$$\tan(x) + 7\sec(x) + 7\cos(x) + C$$

## 34 Problem 35

### 34.1 Problem

$$\int \frac{8}{3x^3 + 7x^2 + 4x}$$

### 34.2 Solution

$$\ln(x^2) + \ln|(3x + 4)^6| - \ln((x + 1)^8) + C$$

## 35 Problem 36

### 35.1 Problem

$$\int \frac{8x}{x^3 - x^2}dx$$

### 35.2 Solution

$$-\ln(x^8) + \ln((x - 1)^8) + C$$

## 36 Problem 37

### 36.1 Problem

$$\int \frac{1}{e^x \sqrt{e^{2x} - 9}} dx$$

### 36.2 Solution

$$\frac{\sqrt{e^{2x} - 9}}{9e^x} + C$$

## 37 Problem 38

### 37.1 Problem

$$\int (\sqrt{x} - x^{-1})^2 dx$$

### 37.2 Solution

$$\frac{x^2}{2} - 4x^{1/2} - \frac{1}{x} + C$$

## 38 Problem 39

### 38.1 Problem

$$\int \frac{e^{\sqrt{3}} + x^{\sqrt{2}}}{x^{1/2}} dx$$

### 38.2 Solution

$$\frac{2x^{1/2+\sqrt{2}}}{1+2\sqrt{2}} + 2e^{\sqrt{3}}x^{1/2} + C$$

## 39 Problem 40

### 39.1 Problem

$$\int \frac{e^{x/3}}{(34 - e^{x/3})^{5/2}} dx$$

### 39.2 Solution

$$\frac{2}{(34 - e^{x/3})^{3/2}} + C$$

## 40 Problem 41

### 40.1 Problem

$$\int \tan(x) \ln(\cos(x)) dx$$

### 40.2 Solution

$$\frac{-\ln^2(\cos(x))}{2} + C$$

## 41 Problem 42

### 41.1 Problem

$$\int \sqrt{25 - x^2} dx$$

### 41.2 Solution

$$\frac{x}{2} \sqrt{25 - x^2} + \frac{25}{2} \sin^{-1}\left(\frac{x}{5}\right) + C$$

## 42 Problem 43

### 42.1 Problem

$$\int \ln^2(x) dx$$

### 42.2 Solution

$$x \ln^2(x) - 2x \ln(x) + 2x + C$$

## 43 Problem 44

### 43.1 Problem

$$\int \frac{\ln(x)}{x^4} dx$$

### 43.2 Solution

$$\int \ln(x) x^{-4} = \frac{\ln(x) x^{-3}}{-3} - \frac{x^{-3}}{9} + C$$

## 44 Problem 45

### 44.1 Problem

$$\int e^{6x} \sin(e^{6x}) dx$$

### 44.2 Solution

$$\frac{-\cos(e^{6x})}{6} + C$$

## 45 Problem 46

### 45.1 Problem

$$\int \frac{10}{(x-1)(x^2+9)} dx$$

### 45.2 Solution

$$\ln|x-1| - \frac{1}{2} \ln(x^2+9) - \frac{1}{3} \tan^{-1}\left(\frac{x}{3}\right) + C$$